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Starvation in Man

G. F. CAHILL JR

In creatures in whom mobility plays an important role in survival, the five to tenfold greater efficiency of storage of excess calories as fat instead of protein or carbohydrate is crucial. This is particularly true for terrestrial and avian species who must struggle continuously against gravity. Thus in man, who once survived as a hunter and gatherer, and in whom mobility also played an important role in his competition with other creatures and, unfortunately, frequently against fellow man, storage of calories as triglyceride in adipose tissue was mandatory for survival. Thus, as discussed elsewhere in this issue, as man eats, the first priority is to provide fuel for immediate metabolic requirements. The second is to expand his modest glycogen reserves in liver and muscle and also to replace the amount of protein broken down in various tissues since the last meal, particularly in muscle. The third priority is to convert the excess, i.e. the originally ingested calories as carbohydrate, protein or fat, into triglyceride and to store the calories in adipose tissue.

In fasting, the priorities are reversed. The body undergoes a series of hormonal and metabolic changes to draw selectively on its extensive supply of calories in adipose tissue, and to spare the breakdown of vitally needed proteins, such as those involved in muscle contractility or in enzymes in critical structures such as heart or liver, or, even more important, proteins involved in nervous tissues, particularly brain, which appear not to be mobilised at all during starvation. This chapter will discuss these various adaptations to a deficit caloric economy, and will describe how the human body progressively and preferentially uses fat for energy.

ENERGY STORES

First, an overall account of calories in various forms and tissues is necessary to provide perspective to fuel economy and mobilisation in man (Table 1). A normal adult uses 1 to 1.2 Cal/min to maintain basal energy needs, or 1500 to 1800 Cal/day. With standard physical activity or in a cold environment, this is doubled, and with strong physical activity, daily expenditures may increase to 5000 to 6000 Cal/day. Thus the 100 000 to 150 000 Calories in the 12 to 16 kg of triglyceride in fat provides one to three months of survival fuel

depending on physical activity, again providing that the supply of calories is rate-limiting in survival, and most evidence suggests that it is (Apfelbaum, 1975). In a very fat human, survival would be much longer, and total starvation of a year has been documented in some very obese subjects under carefully observed conditions in which cheating would be most difficult. In fact, the excretions of nitrogen, electrolytes and ketoacids in these individuals was carefully monitored to police the total or, at least, near total starvation. However, spurious ingestion of 10 to 20 Calories daily or less might not have been detected.

Table 1. *Available fuel reserves in man*

	Glycogen or glucose		'Mobilisable' protein		Triglyceride	
	g	calories	g	calories	g	calories
Blood	15	60	0	0	5	45
Liver	100	400	100	400	50	450
Brain	2	8	0	0	0	0
Muscle	300	1200	6000	24 000	50	450
Adipose	20	80	10	40	15 000	135 000

PHASES OF STARVATION

In the transition from the fed to the fasted state, a sequence of metabolic alterations occurs, listed as follows with their approximate duration:

1. Gastrointestinal absorption of substrate — 1—6 hours
2. Glycogenolysis — 1—2 days
3. Gluconeogenesis — first week
4. Ketosis — 3—4 days onward
5. Diminishing gluconeogenesis and increasing cerebral ketone consumption — second week onward

Gastrointestinal phase

Although this is beyond the general scope of this chapter, a few words on the disposition of a meal is appropriate, especially concerning the variations in caloric content and the distribution in carbohydrate, protein and fat. With a large, mainly carbohydrate, meal the entire body literally metabolises glucose. This is a result of two phenomena, both involving insulin. Liver actively removes glucose due to the elevation in insulin levels, and the suppression of glucagon levels. The glucose is incorporated into glycogen and is also glycolysed to pyruvate and lactate, and the pyruvate subsequently oxidised to acetyl CoA which is used for both liver energy needs via the tricarboxylic acid cycle, and also for fatty acid synthesis. Fatty acid is subsequently incorporated into triglyceride and then the fat is exported to adipose tissue as very low density lipoprotein (VLDL). Brain uses glucose for fuel, as it did before the meal. Muscle, again thanks to the elevated insulin level, preferentially utilises glucose to replenish its own glycogen reserves as well as fuel for energy. This preferential glucose metabolism in muscle is also

a function of low levels of free fatty acids. In adipose tissue, glucose uptake is also stimulated by insulin as is its conversion to fatty acid and subsequent incorporation into the triglyceride vacuole in the centre of the adipocyte. Triglyceride hydrolysis (lipolysis) of the stored triglyceride is also inhibited by insulin, resulting in a decreased release of free fatty acid, and in a lower circulating level of free fatty acid. The larger the meal, the more rapid the rate of glucose uptake by these various tissues, again owing to the higher level of insulin.

The ingested fats enter into the bloodstream as chylomicrons via the lymphatics, and the simple sugars and amino acids are absorbed into the portal circulation going to the liver. The glucogenic amino acids, as well as certain essential amino acids like tryptophan, are removed and metabolised by the liver. The three branched-chain amino acids, leucine, isoleucine, and valine, which will be discussed in further detail later, are mainly removed by muscle and adipose tissues. Thanks to insulin and the increase in amino acid levels, peripheral protein, particularly in muscle, is replenished. The chylomicrons serve to transport absorbed fat directly to adipose tissue where the fatty acids are released, and then incorporated as a new triglyceride molecule and stored inside the triglyceride vacuole in the centre of the adipocyte along with the newly synthesised fatty acids.

Should the meal be low in carbohydrate content, so that the rate of carbohydrate entry into the bloodstream is less than that needed by brain and other obligatory glucose users, insulin is still released at a greater rate than basal in order to initiate peripheral protein synthesis, again especially in muscles, and also to stimulate lipogenesis in adipose tissue from amino acids (particularly the branched-chain amino acids) and uptake of circulating triglyceride into adipose tissue. However, glucose levels must be maintained in the presence of this increased insulin, and thus the liver must be poised towards glucose production in spite of the increased insulin levels. This appears to be the primary biological function of glucagon (Lefebvre and Unger, 1972), whose release is stimulated by both the increase in amino acids themselves, particularly arginine, as well as sensitisation of the alpha cells to release more glucagon per change in substrate level. This is probably mediated by pancreozymin, which appears to be the gut hormone which stimulates not only exocrine pancreatic activity, but also the endocrine alpha and beta cells, the former to release glucagon and the latter to release insulin after protein ingestion.

Thus in meals containing little or no carbohydrate, by initiating increased insulin activity as well as that of glucagon, the non-hepatic tissues receive the 'fed' signal to take up circulating fuels, but the liver remains in the 'fasting' mode in order to maintain blood glucose concentration by continuing to produce glucose, and this it does owing to the increase in glucagon levels. Thus, whether liver is glucogenic or glycogenic—glycolytic is a function of the ratio of the two hormones, insulin and glucagon, as originally postulated by Unger and subsequently corroborated by a number of investigators. If the meal contains sufficient carbohydrate to be able to displace the need for hepatic glucose production — in other words, enough carbohydrate to enter into blood at a rate of over 100 to 200 mg/min — then the slight rise in

glucose concentration is sufficient to increase beta cell insulin release and to suppress the alpha cell from releasing glucagon, and as a result, hepatic glucose production is suppressed. In addition, this slight increase in glucose level markedly synergises the beta cells to produce even more insulin as a response to the increase in amino acids, so the glucagon/insulin ratio is even more altered.

In summary, the gastrointestinal phase of starvation varies as a function of the fuel ingested, and if deficient in carbohydrate, the liver is signalled to produce glucose as if no meal had been eaten. Should this type of diet be continued for periods of time, experimental animal data have shown that the liver becomes more and more poised toward gluconeogenesis due to the increased activities of the key rate-limiting enzymes, such as phosphoenolpyruvate carboxykinase, and the state of the liver is thus metabolically similar to that in total starvation.

Glycogenolysis phase

Total body free glucose (Table 1) is only 15 grammes or one hour's worth of fuel for basal energy needs. If one permits a 'physiological excursion' from the fasting level of 80 mg/100 ml in plasma to 60 mg/100 ml, that would be 15 minutes fuel for the whole body, or if it was limited to the brain's needs, about 45 to 50 minutes. Thus, there must be exquisitely sensitive mechanisms to respond to small changes in glucose concentrations which can result in increased or decreased rates of glucose production by liver (Cahill, 1971). Insulin and glucagon play central roles, and there is also evidence that the glucose molecule itself may be important in liver in controlling its own destiny. First, the liver cell membrane does have a specific glucose transporting mechanism, but its rate (Vmax) is so great that the concentration of glucose inside the liver cell closely approximates that in the circulation. Second, the kinase in liver responsible for forming glucose-6-phosphate is unique in that it has a low affinity for glucose, meaning the higher the glucose level, the more it is phosphorylated. This is different from the enzyme in muscle and adipose tissue, where the rate-limiting step is glucose entry across the membrane into the cell, and once the glucose molecule has entered, owing to insulin's effect on the membrane, it is rapidly phosphorylated because of the very high affinity (low Km) that the hexokinase in muscle and adipose tissue has for glucose. Thus, in liver, hyperglycaemia results in increased glucose phosphorylation, and thereby in glucose uptake, simply as a function of its concentration.

To return to glucagon and insulin, the ratio of these two hormones plays a central role in controlling the level of cyclic AMP inside the liver cell. Both hormones bind to specific receptors on the liver cell membrane. Glucagon activates the formation of cyclic AMP (CAMP) and insulin is thought to increase its rate of destruction, the two enzymes for synthesis and breakdown being adenyl cyclase and phosphodiesterase, respectively. In the liver cell, elevated levels of CAMP initiate one or more enzymes which phosphorylate various proteins, and these result in inactivation of enzymes responsible for glycogen synthesis and activation of those involved in glycogenolysis and gluconeogenesis. The entire system serves as a mechanism whereby a few

glucagon molecules can literally turn off glycogen synthesis within seconds and can, by a waterfall cascade with amplification at each step, turn on millions of enzyme molecules to rapidly break down glycogen. Other slower effects initiated by an increase in CAMP in liver decrease the activity of enzymes of other pathways of glucose utilisation, such as glucokinase, and increase the activities of enzymes involved in fat breakdown and in gluconeogenesis, such as phosphoenolpyruvate carboxykinase. CAMP has been suggested to serve in nature as a signal for lack of carbohydrate (Tomkins, 1975), and this seems to be particularly applicable to man.

As glucose absorption decreases from the gut at the end of the gastrointestinal phase, insulin levels also fall, and the liver gradually stops removing glucose; it then is more or less in 'neutral' and subsequently begins to produce glucose. Most clinicians who follow glucose levels in venous blood during the normal glucose tolerance test feel the meal is absorbed in two to three hours, as evidenced by normal venous glucose levels, but it is probably much longer, depending on its size. The mechanism for this normoglycaemia is probably the release of one or more gastrointestinal hormones as carbohydrate is absorbed (Creutzfeldt, 1974). This may be due to newly described peptides such as VIP (vasoactive inhibitory peptide) (Schebalin et al, 1974) or GIP (gastrointestinal inhibiting peptide) (Dupre, Watson and Brown, 1973) which facilitate a greater insulin release per unit change in glucose. As long as carbohydrate is being absorbed into blood, it is being removed by the tissues at the same rate, with only a minimal increase in arterial levels, again owing to the high insulin levels. Venous glucose levels are usually at the pre-fasted level after the first hour or two, and in many cases, may even be below this while the meal is still being absorbed (the 'flat' glucose tolerance as seen in many healthy individuals).

About four to five hours after a meal, and perhaps longer if it was a very large meal, the liver begins to return its stored glycogen as free glucose back into the blood to provide the fuel needs mainly for the central nervous system (Figure 1). The signals are two-fold: a lower insulin level and a lower level of portal blood glucose. Whether glucagon, no longer suppressed, increases or not in this brief time is not yet clearly documented. It is the author's guess that the 'fine-tuning' is primarily a function of insulin and glucose, and not glucagon. In experiments in fasted humans, doubling the glucagon concentration by infusion had little effect on glucose homeostasis (Sherwin and Felig, 1976). However, suppression of both glucagon and insulin by somatostatin lowers blood glucose (Gerich et al, 1975) suggesting the presence of glucagon to be necessary for hepatic glucose production, but not regulatory.

During this phase of decreasing glucose and insulin levels, peripheral tissues such as muscle and adipose progressively diminish glucose utilisation, so that after eight to ten hours, over half of muscle fuel needs are met by free fatty acid oxidation. Simultaneously, the levels of free fatty acids rise as the insulin levels fall, primarily due to increased levels of CAMP stimulating triglyceride lipolysis. Whereas in liver insulin is pitted against glucagon in controlling CAMP concentration, in adipose tissue it is insulin versus nor-epinephrine released from sympathetic nerve endings. Knowledge of the

contribution of muscle glycogen to fuel utilisation as a direct result of fasting without exercise is limited. Human muscle after a meal may be as high as one per cent glycogen. With exercise it is rapidly diminished and, if fasting persists, it is not replenished. With the 30 kg of muscle in a normal man, one per cent glycogen would be 300 g or 1200 Calories (Table 1), a nominal supply, possibly explaining the persistence of a higher respiratory quotient in starving man than can be explained solely by release and oxidation of the glucose stored as glycogen in liver.

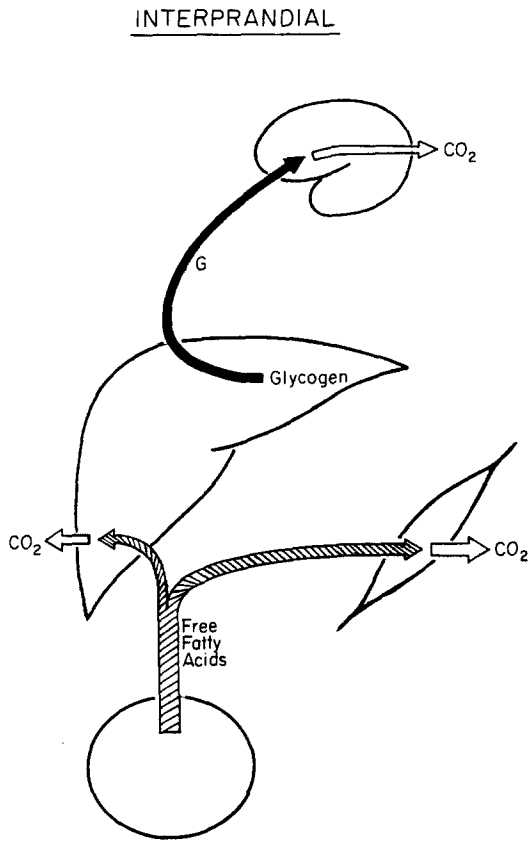

Figure 1. Between meals, glucose for brain is provided by hepatic glycogenolysis, and free fatty acids released from adipose tissue become the predominant fuel for muscle. Low insulin levels promote the hepatic glycogenolysis, the diminished glucose entry into muscle, and the increased level of free fatty acids.

Gluconeogenesis

Although there are few data concerning liver glycogen levels after a large meal, extrapolating backward from the four to five per cent in liver in the post-absorptive state, there may be as much as ten per cent or more. Thus

liver glycogen maintains blood glucose for 12 to 16 hours, and studies by Felig and Wahren (Felig, 1975) in the post-absorptive state show glycogen providing three-quarters of splanchnic glucose output and gluconeogenesis the remainder (Figure 2).

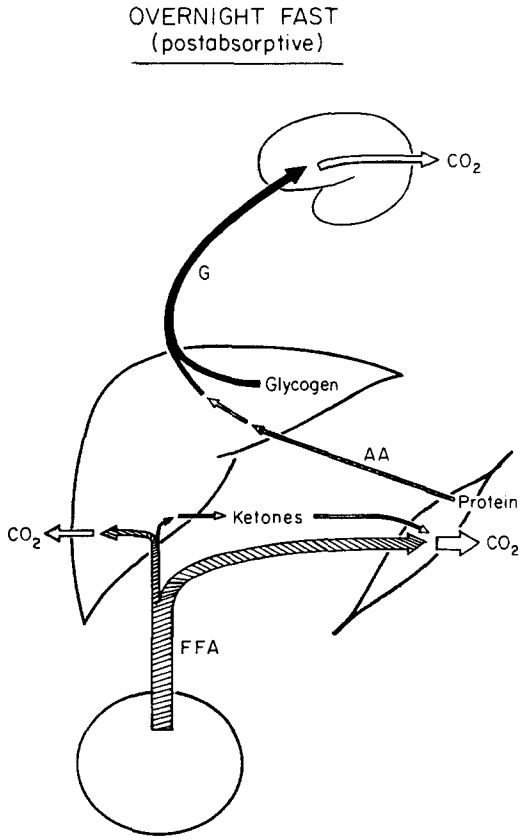

Figure 2. With a fast of over 12 hours, as liver glycogen becomes depleted, gluconeogenesis from amino acids originating in muscle provides glucose for brain. Ketoacid production by liver becomes more prominent.

For gluconeogenesis to be initiated, a number of metabolic changes must occur in the liver. In experimental animals, glucokinase activity is decreased and glucose-6-phosphatase, its opposing enzyme, is increased. Also the liver glycogenolysis system out-balances the synthetase system (Figure 3). At the level of 3-carbon intermediates, pyruvate is no longer oxidised to acetyl CoA for further oxidation or incorporation into fatty acids. Instead, the pyruvate is carboxylated to oxaloacetate, and this in turn is converted to phosphoenol-pyruvate which is then swept upward into glucose by the higher activity of another controlling stage, fructose-1, 6-diphosphatase, which is increased in activity, and phosphofructokinase, which is decreased. All of these alterations

are a result of two processes, a higher glucagon/insulin ratio increasing levels of CAMP due to a significant decrease mainly in insulin, and the second, a higher level of free fatty acids, which increases fat oxidation and produces a higher level of fat-derived materials such as acetyl CoA and fatty acyl CoA. Other effects of fat oxidation are a lowering of levels of NAD^+ and free CoA, cofactors needed for pyruvate to be oxidised. As these enzymes and cofactors all change over 12 to 24 hours, the rate of liver glucose output begins to be controlled by the level of substrate coming to it, and the rate-control is thus transferred from liver to peripheral tissues, as suggested by the studies of Mallette, Exton and Park (1969) and Felig et al (1968).

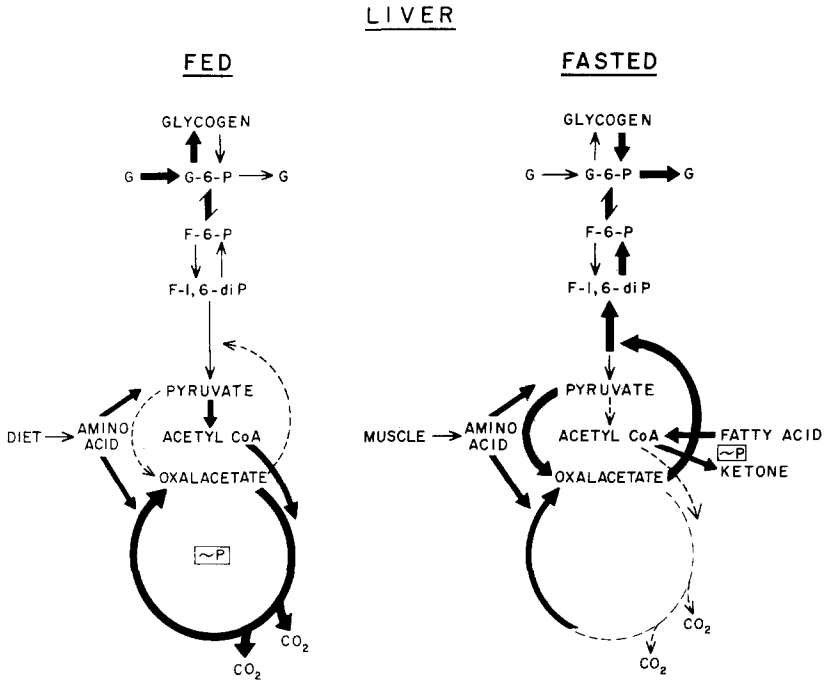

Figure 3. Pathways of metabolism in liver in the fed (low glucagon/high insulin) and fasted (high glucagon/low insulin) states. In the fed state glycogen and fatty acid synthesis predominate and energy is derived from the tricarboxylic acid cycle. In the fasted state, glycogenolysis, gluconeogenesis and ketogenesis predominate, the latter to provide energy for liver's needs.

Over the ensuing two to three days of starvation, muscle and adipose tissue become progressively more efficient in preventing glucose utilisation, both by blocking glucose uptake and, as a further check, by preventing glycolysis of the glucose to pyruvate and, as a final check, completely preventing pyruvate from being oxidised to acetyl CoA, so what little pyruvate is formed is either exported as such or is reduced to lactate and exported, or is transaminated to alanine and then sent back to the liver for gluconeogenesis (Aoki et al, 1975b).

Returning to the liver, its own energy needs are met by oxidation of free fatty acids, but as starvation progresses, and the oxaloacetate is utilised more for gluconeogenesis, less is available for tricarboxylic acid cycle activity. Furthermore, what oxaloacetate is available is reduced to malate as the liver is flooded with free fatty acids and with the reducing equivalents resulting from their oxidation. Thus, as the fatty acids are dehydrogenated and split into two carbon units of acetyl CoA, the diminished acceptor, oxaloacetate, decreases acetyl CoA entry into the cycle and the liver's alternative is to export the acetyl CoA, two molecules at a time, as β-hydroxybutyrate or acetoacetate, the 'ketone' bodies. Ketone production appears to be the result of two phenomena, the high glucagon/insulin ratio, which increases the enzymatic machinery for ketone production, and an increase in the delivery of free fatty acids from adipose tissue (Grey, Karl and Kipnis, 1975).

Thus, to summarise the gluconeogenic phase, liver enzymes become poised toward glucose synthesis, fat becomes the hepatic fuel, but the fatty acids are only partly oxidised as far as acetate units, and then exported to the periphery two at a time as acetoacetate and β-OH butyrate. Adipose tissue hydrolyses its triglyceride to free fatty acids and glycerol (lipolysis), the former serving as fuel for muscle and liver, and the latter, like amino acids from muscle, or like lactate and pyruvate from muscle and other tissues, is used by liver as gluconeogenic substrate.

Quantitatively, estimated 24-hour substrate turnover in a normal man fasted for three to four days is approximately:

Brain glucose utilisation	100 grammes
Brain ketone utilisation	50 ,,
Splanchnic glucose output (Felig et al, 1969)	150 ,,
Glucose utilisation by other tissues	50 ,,
Muscle proteolysis (Cahill et al, 1966)	75 ,,
Adipose lipolysis	180 ,,
Splanchnic ketogenesis (Reichard, Owen and Haff, 1974)	150 ,,

Thus the brain is beginning to utilise ketoacids as their levels increase in blood; glucose utilisation begins to diminish; splanchnic ketoacid production is maximal and the bulk of ketoacids is utilised by peripheral muscle and heart. In fact, the heart has been shown to satisfy three-quarters of its energy needs by ketoacid metabolism in a three-day fasted man. Ketoacid loss in the urine is minimal calorically, but due to their acidic nature, necessitates ammonia production by kidney to prevent loss of cation and thus to preserve body fluid volume (Sapir and Owen, 1975).

Ketosis

As mentioned above, by the third day of starvation, ketoacid production by splanchnic bed is maximal (Garber et al, 1974), but blood levels continue to increase progressively until the end of the second week when a plateau is achieved (Cahill et al, 1966). Owen and Reichard (1971) have shown that this progressive increase is mainly a function of decreased ketoacid metabolism

by muscle as fasting progresses. Thus by the third or fourth day, ketoacid levels are 1 to 2 mmol/l but by the second week may be 6 to 10 mmol/l. Serum bicarbonate levels are reduced accordingly, and there is a mild but compensated metabolic acidosis. The purpose of this ketoacidosis is to provide a sufficient gradient for the facilitated diffusion across the blood—brain barrier to satisfy the brain's energy needs (Owen et al, 1967), Again, this production of ketoacid by liver is a result of a high glucagon/insulin ratio and an increase in the levels of free fatty acids released from adipose tissue, another function of low insulin.

Diminishing gluconeogenesis and increasing marked ketoacid consumption

The final phase in starvation occurs as ketoacid levels plateau and the brain is preferentially using ketoacids as fuel and diminishing glucose utilisation accordingly (Figure 4). The net effect is to decrease the need for gluconeo-

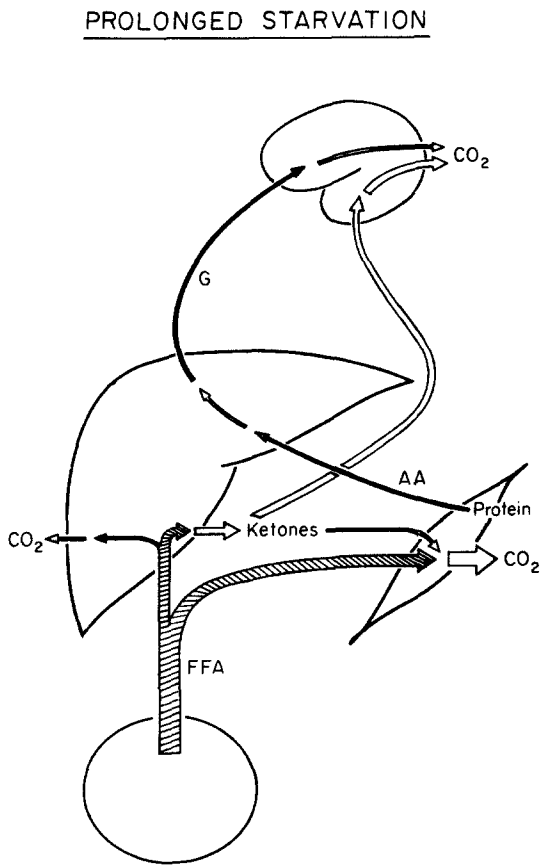

Figure 4. Flow of substrate in prolonged fasted man with ketoacid utilisation by brain, diminished ketoacid utilisation by muscle, and diminished glucose utilisation by brain resulting in decreased daily protein catabolism. The glucose still used by brain is that derived from glycerol from adipose triglyceride plus that from amino acids derived from muscle and metabolised by liver and kidney into glucose.

genesis and thus to spare mobilisation of muscle protein. Nitrogen excretion diminishes from 12 g/24 hours to 3 to 4 g/24 hours, signifying a reduction of protein catabolism from 75 g to 12 to 20 g daily (Figure 5). Thus starvation can continue utilising the plentiful fat calories and sparing necessary protein. Insulin levels continue to be low, but insulin appears to be the controlling signal for integrating the various metabolic processes. Glucagon appears to return to post-absorptive levels, but the glucagon/insulin ratio is still in favour of glucagon, so the liver is poised in the direction of gluconeogenesis.

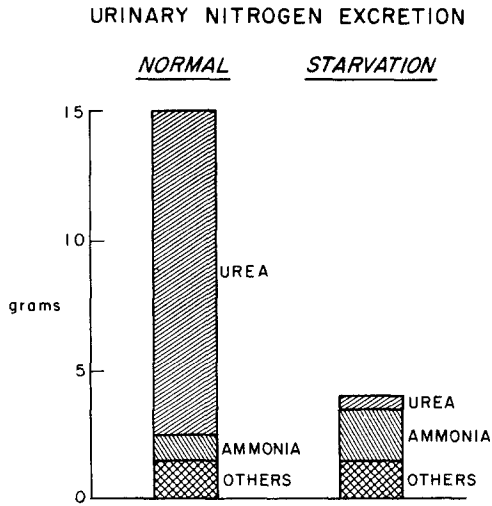

Figure 5. Urine nitrogen excretion of the average man on approximately 100 g protein intake showing the predominance of urea, and in a man after several weeks of starvation showing the small urea excretion and the predominance of ammonium.

Quantitative estimates of 24-hour substrate turnovers are estimated as:

Brain glucose utilisation	40	grammes
Brain ketone utilisation (Owen et al, 1967)	100	,,
Splanchnic glucose output (Owen et al, 1969)	80	,,
(a) from amino acid	20	,,
(b) from glycerol	20	,,
(c) from returning lactate and pyruvate	40	,,
Muscle proteolysis	20	,,
Adipose lipolysis	180	,,
Splanchnic ketogenesis (Reichard, Owen and Haff, 1974)	150	,,

ROLE OF MUSCLE

With starvation, as mentioned earlier, the overall rate-control for gluconeogenesis is switched from liver to muscle, or, to paraphrase, liver becomes substrate-regulated by muscle proteolysis. As shown over 50 years ago by VanSlyke, amino acids respond to insulin. Wool and Cavicchi (1967) and others have shown a very important role of insulin in promoting and maintaining the protein synthetic machinery inside muscle. Thus, a major effect of insulin is to initiate uptake of certain amino acids and their incorporation into protein. A more recently described effect of insulin is to decrease muscle proteolysis, as shown by several workers including Jefferson et al (1974) and Fulks, Li and Goldberg (1975). Thus, insulin appears to control the rate of net muscle proteolysis — the lower the insulin, the more rapid the net proteolysis. However, as discussed above, muscle proteolysis diminishes with more prolonged starvation and yet insulin levels are low; thus we have a paradox.

Much emphasis has been recently directed to the role of amino acids in intermediary metabolism in man. In the post-absorptive state and in prolonged starvation, alanine and glutamine are released in greater quantities than explicable by their contents in muscle protein (Marliss et al, 1971). Apparently there is a rearrangement of amino acids resulting from proteolysis with the branched-chain amino acids transaminating to pyruvate to form alanine or donating ammonia for glutamine synthesis. Other amino acids contribute carbons to form the pyruvate or α-ketoglutarate for the formation of alanine, or glutamate and glutamine. Some of the glutamate for glutamine formation is taken up from the blood as it courses through muscle. The preferential release of alanine and glutamine is teleologically sound, since alanine is the preferred gluconeogenic substrate for liver and glutamine for kidney. Some of the glutamine is made into alanine by the non-hepatic splanchnic bed, so alanine becomes an even greater hepatic gluconeogenic precursor (Felig, 1975).

The explanation appears to lie in the unique interrelationships between keto-acids, fatty acids and the metabolism of the branched chain amino acids in muscle mitochondria. As was previously mentioned, early in starvation ketoacids serve as muscle fuel, but with more prolonged starvation they are less well used. In fact, as acetoacetate is taken up by the muscle, it is returned back to the blood as β-hydroxybutyrate, signifying a more reduced state of the muscle mitochondria — secondary to free fatty acid utilisation. Thus fatty acids appear to take preference for oxidation and ketoacid oxidation ceases, sparing the ketoacids for the brain, a superb survival process. But the excess oxidation of fatty acids also seems to result in the decrease in oxidation of the deaminated residues of the three branched-chain amino acids, leucine, isoleucine and valine (Aoki et al, 1975b). Thus fat spares their oxidation (Figure 6), and they appear to play a central role in maintaining muscle protein, particularly leucine (Buse and Reid, 1975; Fulks, Li and Goldberg, 1975).

Thus the 'nitrogen-sparing' of prolonged starvation really results from fat being selectively used in muscle. One can show in experimental animals that

fat prevents the terminal nitrogen catabolic phase in starvation (Cuendet et al, 1975). As will be discussed later, feeding but a few calories as protein one is able to replace the nitrogen depletion of otherwise total starvation, and this concept has been capitalised by Blackburn and Flatt in their proposal that endogenous fat, if allowed to be mobilised by not giving carbohydrate, may even be more efficient in sparing nitrogen in patients receiving parenteral alimentation (Blackburn et al, 1973).

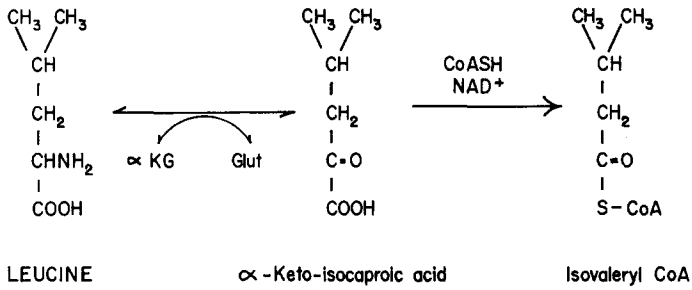

Figure 6. Leucine catabolism in muscle. The cofactors for oxidation of its ketoanalogue, keto-isocaproic acid, are shared with fatty acid oxidation. Thus fatty acid utilisation spares the irreversible conversion of the keto-isocaproic acid to isovaleryl CoA, and results in sparing of leucine, and in turn diminution of muscle protein catabolism.

ROLE OF KIDNEY

Under normal circumstances, kidney in man provides less than ten per cent of glucose production, although few studies have been done. In prolonged starvation, as overall gluconeogenesis decreases, the component provided by liver decreases dramatically and that of kidney increases proportionate to the increase in ketoacid loss in urine, necessitating ammonia excretion to prevent loss of cation in urine, and this in turn is coupled with increased glucose synthesis. After several weeks of starvation, kidney provides up to one-half of total gluconeogenesis, using the same substrates in liver but in different proportions, i.e. amino acids, glycerol, lactate and pyruvate, in that order (Owen et al, 1969). The amino acids, primarily glutamine, are used for ammonia synthesis, and in fact there is more nitrogen in the urine as ammonia than as urea in prolonged starvation. Thus, as far as amino acid-derived gluconeogenesis is concerned, the kidney is even more important than the liver.

CLINICAL PHENOMENA IN STARVATION IN MAN

Weight loss

During the gluconeogenic phase, up to one pound of lean flesh may be lost daily in addition to the one-third to one-half a pound of fat, total tissue weight loss approximating to one to one and a half pounds. However, a total weight loss is far greater, unless the subject had previously been on a restricted carbohydrate intake. This is due to a saline diuresis, the precise mechanism of which has yet to be adequately explained. Some evidence suggests that it

may be related to an increase in the glucagon/insulin ratio (De Fronzo et al, 1975; Spark et al, 1975). In grossly overweight individuals, this diuresis may result in 10, 20 or even 30 pounds or more of weight loss. In normal, non-obese subjects, four to six pounds may be lost, particularly females. Administration of NaCl, or of adrenal mineralocorticoids has essentially no effect (Spark et al, 1975). Fluid balance is exquisitely sensitive to intake of even small amounts of carbohydrate and on refeeding a previously fasted individual, sodium excretion may fall to less than 1 mmol/day. Thus in some subjects, if placed on a liberal salt intake during refeeding, gross oedema may occur, occasionally even compromising circulatory stability leading to congestive failure in those with heart disease.

Later in total starvation, after the gluconeogenic phase and the saline diuresis, weight loss falls to what one would calculate, a quarter of a pound of lean tissue and a third to half a pound of fat daily, for a total of approximately one pound daily.

Brain function

It is frequently asked how well the brain functions when using predominantly ketoacids as fuel. Intellectual function appears to remain intact, but emotional alterations are noted, such as depression or lability. Most individuals decrease spontaneous activity, obviously as a mechanism to spare calories, but physical activity can be marked, as in one marathon runner whose pre-race preparation was total calorie abstention for one week!

Of much interest is the elevation of the electro-convulsive threshold of the brain utilising ketoacids as fuel. In the 1920s this was initiated as a form of therapy in children with seizure disorders and has again been resurrected for those children in whom use of multiple anticonvulsive agents is yet ineffective.

The hypothalamic area appears to be significantly altered in the ketosis of prolonged starvation. Appetite is diminished and this may be part of the success of the ketogenic diets used in the treatment of obesity. More dramatically, the desire for fluid intake is diminished, and fasting subjects usually need to be encouraged to drink water. On this point, the markedly diminished urea excretion diminishes the osmotic excretory load, and urine output of 100 to 200 ml/day may be all that is necessary. Again, the survival value is obvious; a fasting man need drink very little water, and the 200 to 300 ml/day produced by metabolism may permit him to survive water deficits for literally several days to a week or more, providing he is not in a hot, dry, dehydrating climate (Cahill and Owen, 1968).

Of greater interest, libido is markedly decreased in starvation, and there are significant reductions in the release of pituitary gonadotropins, particularly FSH (Newmark et al, 1976). Thus in the female there is first anovulation and subsequently amenorrhoea. In the male there is decreased testicular function. These changes are similar to those noted in subjects with anorexia nervosa (Boyar et al, 1974).

Other metabolic sequelae

With progressive ketosis and the mild metabolic acidosis, there is a gradual and continual excretion of calcium and phosphorous in amounts beyond that

lost from the lean tissue being catabolised. Thus, bone mineral is gradually being dissolved, similar to that noted in chronic renal acidosis, but to a much lesser degree. This phenomenon has made some clinicians reluctant to use starvation or ketogenic diets in individuals prone to osteoporosis, such as Caucasian females, for prolonged periods of time.

More significantly, as ketosis becomes moderate, the kidney retains uric acid to a greater degree, and serum urate levels rise to 8 to 10 mg/100 ml, and in some individuals to levels as high as 15 to 18 mg/100 ml. Amazingly, episodes of gout are rare in spite of this super-saturation, and the attacks that have been precipitated have been in individuals with previous disease.

In rare subjects, hepatic decompensation may occur, first evidenced by liver enlargement which on examination shows the typical changes of fatty infiltration. The mechanism for this is allegedly mobilisation of excess free fatty acids to the liver with deficient synthesis of lipoprotein to provide export back to the periphery. Most of these subjects probably either had hepatic disease or were on an imbalanced diet prior to starting the fast. It is of interest that vitamins are probably unnecessary with total starvation, the need for the B vitamins, particularly B_1, and for vitamin C occurring only with carbohydrate intake. An Arctic explorer eating crackers died with scurvy and beriberi while his Eskimo guide survived on blubber! Probably contributing to the rare hepatic decompensation in starvation is the decreased splanchnic flow (Drenick, 1968), and thus total starvation is contra-indicated in individuals with liver disease. It should be remembered that small increases in bilirubin can occur even in normal fasting people and need not signify hepatic disease (Barrett, 1971).

Metabolic rate

A decrease in metabolic rate has been noted in starvation for decades, having been extensively studied by Dubois, Benedict and others. Part of this is explained by the progressively decreasing lean body mass, but the decrease appears to be more than accounted for by decreased metabolisable mass. The selective use of fat as fuel, everything else being equal, would be expected to increase oxygen consumption, since it is slightly less efficient than glucose as an energy source, yet, as above, total oxygen consumption is decreased about 10 to 15 per cent.

Recently investigators have noted that the levels of thyroxine remain normal during starvation, but those of triiodothyronine decrease strikingly into the hypothyroid range (Portnay et al, 1974; Vagenakis et al, 1975). In contrast, the level of the inactive triiodo form ('reverse' T_3 in endocrine jargon) increases. When re-fed with carbohydrate, a fasted man reverses the levels of the two triiodothyronines and the active form returns to euthyroid levels. Whether the decrease in active 'T_3' is the sole mediator of the decreased oxygen consumption remains to be demonstrated. In any case, administration of triiodothyronine to fasted subjects not only increases oxygen consumption, but also nitrogen excretion (Carter et al, 1975); however, a high protein intake maintains nitrogen balance (Lamki et al, 1973).

PARTIAL STARVATION

CARBOHYDRATE SUPPLEMENTATION. Administration of small amounts of carbohydrate to prolonged fasted man results in a decrease in the level of ketosis and urinary nitrogen excretion (Aoki et al, 1975a). It is interesting that levels of free fatty acids are unaffected, as might be predicted, since the administered glucose (120 g/day) is sufficient just for the brain, ketone levels having fallen dramatically, and the remainder of the body must continue to use free fatty acids as fuel. Total urinary nitrogen decreases from three grammes daily with total starvation to one to two grammes daily with the supplemental carbohydrate. Similar effects were observed by simply infusing insulin; thus the observed metabolic effects are probably secondary to this hormone.

In clinical terms, giving small meals of carbohydrate to totally starved individuals in the amount of 400 to 500 cal/day does decrease net N-loss, but the nitrogen sparing already achieved by the fat oxidation is so efficient that the relative practical economy is small.

PROTEIN OR AMINO ACID SUPPLEMENTATION. As discussed by Apfelbaum elsewhere in this issue, giving totally fasted humans small amounts of protein, as discussed by Folin over 60 years ago, can maintain nitrogen balance and permit selective utilisation of fat calories as fuel. This concept has been used by Blackburn and Flatt (Blackburn et al, 1973; Hoover et al, 1975) as a mode of therapy for subjects unable to eat, such as postoperative patients, in whom total starvation plus added amino acids maintain better nitrogen balance than total starvation, or by administration of equicaloric amounts of glucose without amino acids.

The mechanism whereby this is achieved appears to lie in the branched-chain amino acids which not only are rate-limiting essential precursors to muscle protin synthesis but also, especially leucine, promote the synthesis of protein and decrease its breakdown directly (Buse and Reid, 1975; Fulks, Li and Goldberg, 1975). Proof of their essential role is the ability of totally fasting man to use the non-amino carboxylic acid analogues of the branched-chain amino acids (Sapir et al, 1974) with a resulting significant reduction in nitrogen excretion. Thus, as Apfelbaum has popularised, it is logical to feed small amounts of protein to otherwise fasted man to spare his body nitrogen and yet provide the barest minimum of calories from external sources.

CONCLUSION

In summary, starvation entails a progressive selection of fat as body fuel. Soon after a meal glucose utilisation by muscle ceases and fatty acids are used instead. Ketoacid levels in blood become elevated over the first week, and the brain preferentially uses these instead of glucose. The net effect is to spare protein even further, as glucose utilisation by brain is diminished. Nevertheless, there is still net negative nitrogen balance, but this can be nullified by amino acid or protein supplementation. Insulin appears to be the principal regulatory hormone. Recent data suggest that decreased levels of active T_3 may play a role by sparing otherwise obligated calories by decreasing metabolic needs.

Table 2. *Fuel flux in man*

State	Hormones		Liver			Muscle				Adipose tissue		Brain	
	Insulin μU/ml	Glucagon pg/ml	Glucose uptake	Glucose production: glycogen	Glucose production: gluconeogenesis	Fuel[a]: Glucose	Fuel[a]: Ketone	Fuel[a]: FFA	Amino acid uptake/release	Glucose uptake	FFA release	Glucose uptake	Ketone uptake
Large meal (CHO 100, P 40, F 40)	100	50	++++	0	0	++++	0	0	+++/0	++++	0	++++	0
Intermediate meal (CHO 50, P 30, F 30)	75	50	+++	0	0	+++	0	±	++/0	+++	±	++++	0
Low carb. meal (CHO 20, P 40, F 40)	50	50	±	±	0	±	0	+++	+/0	±	++	++++	0
Protein—fat meal (CHO 5, P 40, F 40)	40	150	0	++	++	0	0	++++	+/0	0	+++	++++	0
Inter-prandial	15	100	0	++++	0	++	0	++	0/0	0	+++	++++	0
Postabsorptive	12	100	0	+++	+	+	0	+++	0/++	0	+++	++++	0
Early fast (3 days)	10	150	0	±	+++	±	++	++	0/+++	0	++++	+++	+
Intermediate fast (7 days)	8	100	0	0	++	0	+	+++	0/++	0	++++	++	++
Prolonged fast (14+ days)	6	100	0	0	+	0	0	++++	0/+	0	++++	+	+++
Diabetic ketoacidosis	<5	300	0	0	++++	0	±	++++	0/++++	0	+++++	+++	±

[a] FFA = free fatty acids.

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